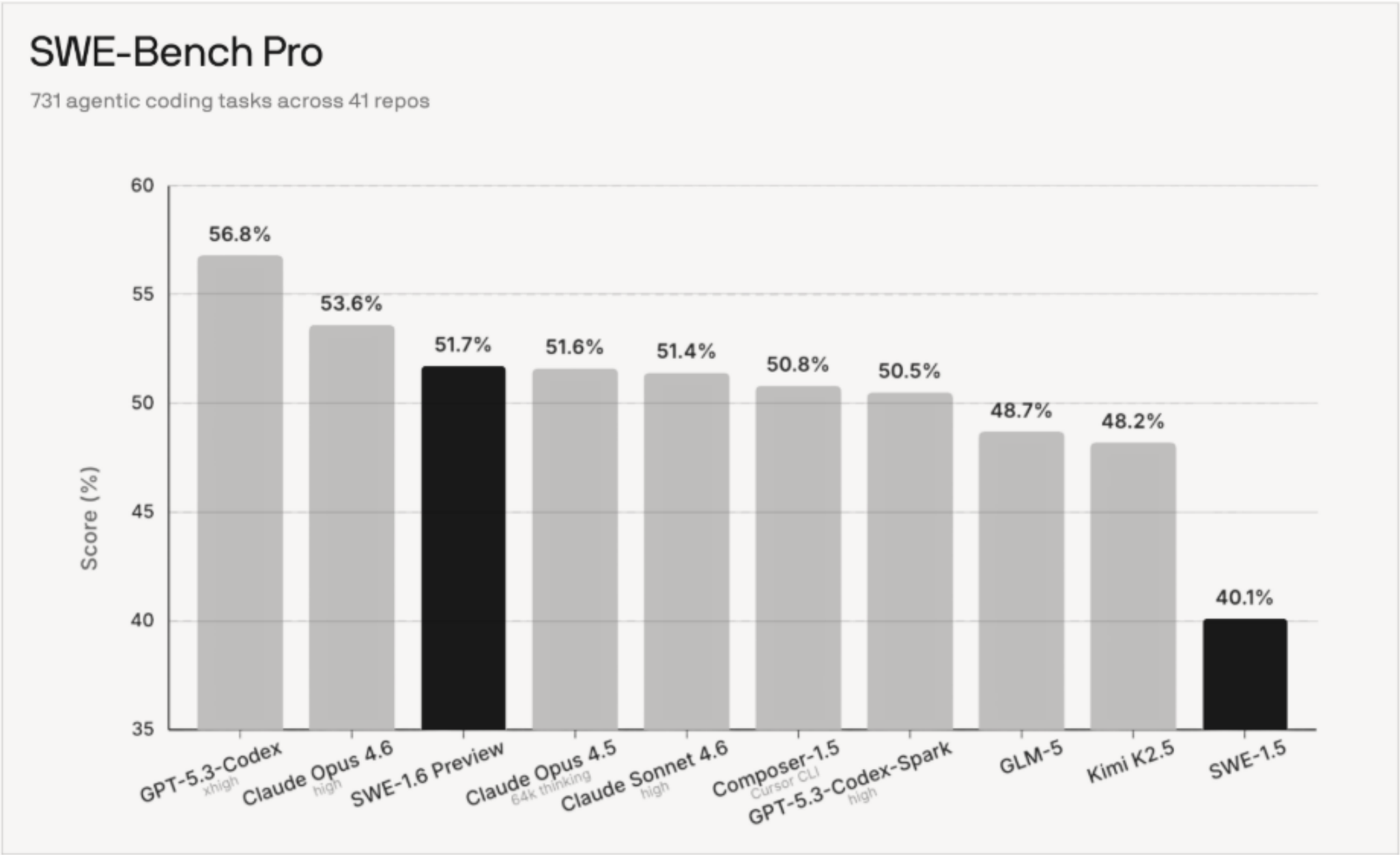
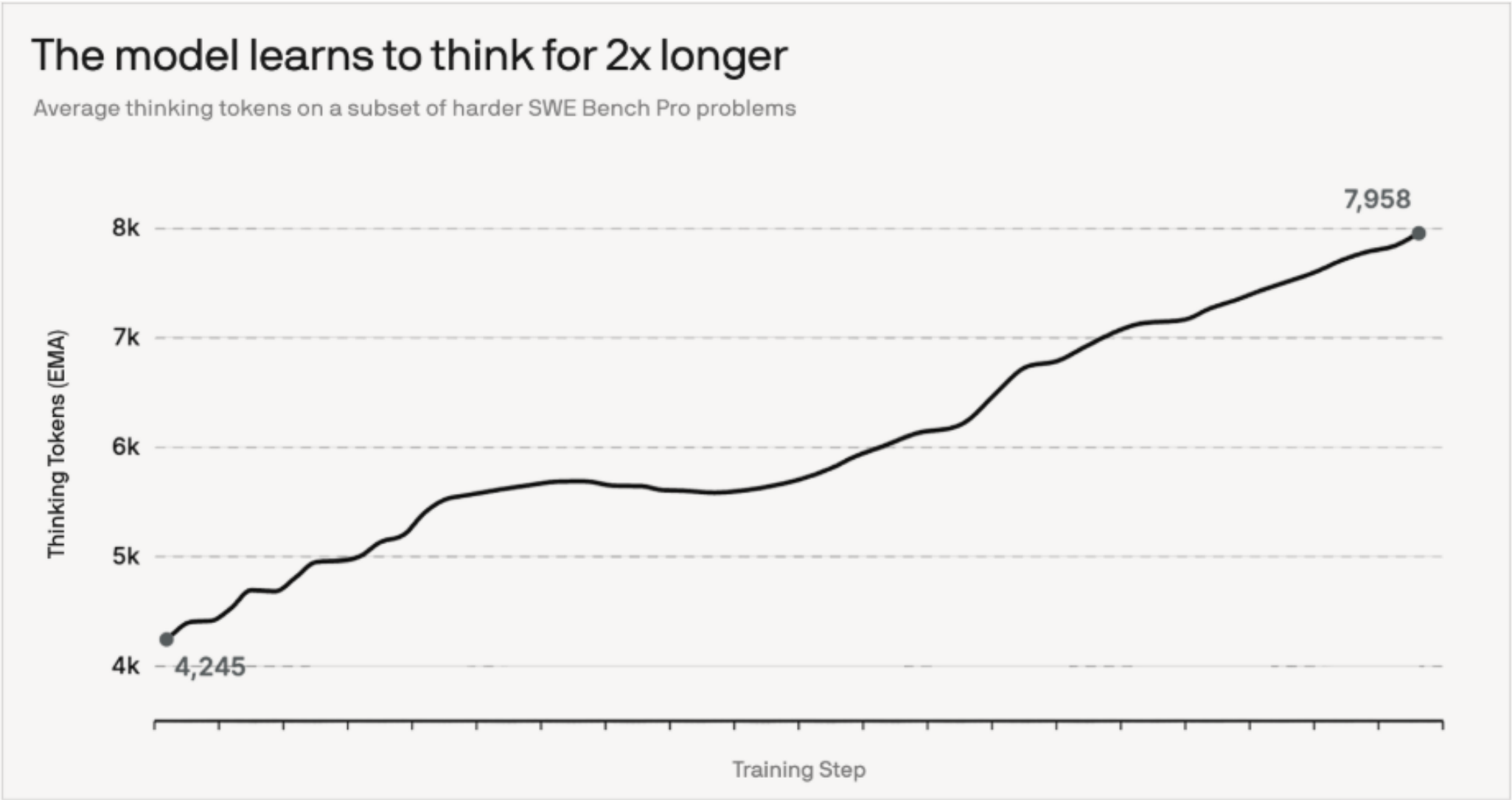




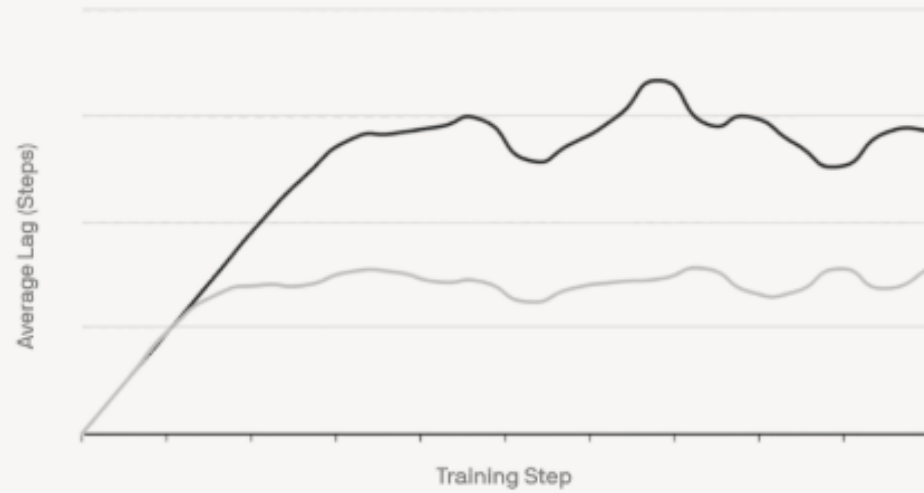




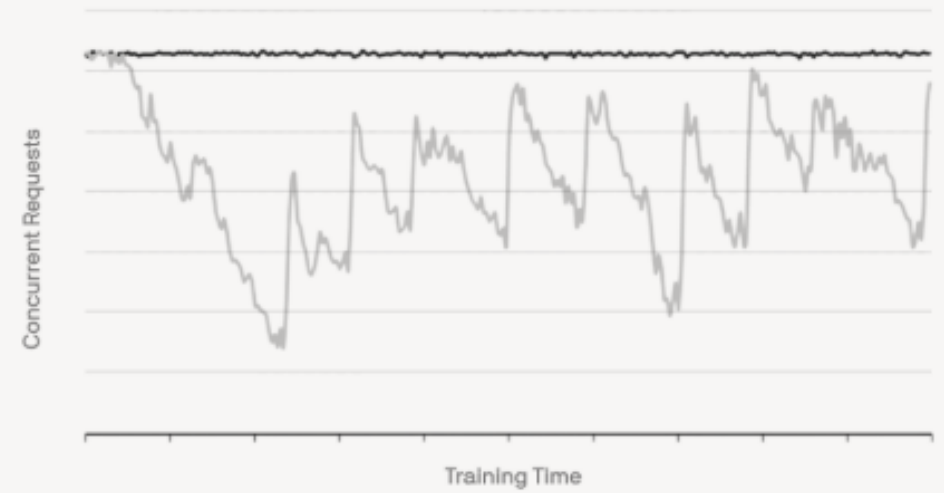
Tolerating higher staleness enables full inference utilization

— Baseline — Higher staleness

Average Staleness of Trajectories



Utilization of Inference Engines



$$t_{roll} \approx \frac{BL_{out}}{n_i s_{roll}}$$

$$t_{train} \approx \frac{BL_{tot}}{n_t s_{train}} = \frac{BL_{out}(1+r)}{n_t s_{train}}$$

$$n_t \approx \frac{N}{1 + \frac{s_{train}}{s_{roll}(1+r)}}, \quad n_i = N - n_t$$

$$\text{avg staleness} \approx \frac{c}{B} \frac{N}{1 + \frac{s_{\text{roll}}}{s_{\text{train}}} (1 + r)}$$

